Thanh-Long V. Le

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Research Interests

My research focuses on **reinforcement learning (RL) post-training for foundation models**, spanning LLM reasoning and generative modeling. My long-term vision is to develop **general-purpose, self-improving foundation models** that can autonomously improve through fine-grained and self-supervised RL, while adapting to unfamiliar tasks through **multimodal in-context understanding and continuous prediction**.

Education

Korea Advanced Institute of Science and Technology (KAIST AI) **Feb. 2025 – Jun. 2027**
M.S. in Artificial Intelligence *Seoul, South Korea*

- GPA: **4.10/4.30**

Korea Advanced Institute of Science and Technology (KAIST) **Sep. 2019 – Feb. 2024**
B.S. in Computer Science and Artificial Intelligence *Daejeon, South Korea*

- Graduated **Magna Cum Laude**
- GPA: **3.98/4.30** (ranked top 8% in School of Computing)

Research Experience

EPFL **Jan. 2027 – May. 2027**
Visiting Researcher, advised by Prof. Maria Brbić *Lausanne, Switzerland*

- Conducting research on AI Agents, World Models, and Video Generation (tentatively).

Adobe Research **Aug. 2025 – Present**
Research Scientist Intern, advised by Dr. Viet Lai *San Jose, CA, USA*

- Led the RL-ZVP project, developing a novel algorithm that enhances LLM reinforcement learning by leveraging learning signals from zero-variance prompts.
- Developed FlowTool, a novel framework combining conditional flow matching and reinforcement learning to efficiently generate continuous tool parameters for image retouching

Machine Learning and Intelligence Lab, KAIST AI **Feb. 2025 – Present**
Gradute Researcher, advised by Prof. Eunho Yang *Seoul, South Korea*

- Conducting research on LLM post-training, including LLM reasoning and alignment through reinforcement learning.
- Currently focusing on augmenting RLVR with various dense learning signal to improve the reasoning ability of LLM.
- Lead a collaborative project with AITRICS on developing a large reasoning model for highly imbalanced clinical event prediction using patient EHR data.

Mobile Intelligence & Interaction Lab, KAIST **May. 2023 – Sep. 2024**
Undergraduate Researcher, advised by Prof. Sung-Ju Lee *Daejeon, South Korea*

- Conducted research on natural language processing, federated learning and human-computer interaction.
- Actively participated in the Exphrase project – an AI-based paraphrasing system with multiple AI-assisted features.
- Addressed label deficiency in federated learning by implementing semi-supervised learning with adaptive thresholding and sharpness-aware regularization.

Publications

No Prompt Left Behind: Exploiting Zero-Variance Prompts in LLM Reinforcement Learning via Entropy-Guided Advantage Shaping
Thanh-Long V. Le, Myeongho Jeon, Kim Vu, Viet Lai, Eunho Yang
*International Conference on Learning Representations (**ICLR**) 2026*

$(FL)^2$: Overcoming Few Labels in Federated Semi-Supervised Learning
Seungjoo Lee, **Thanh-Long V. Le**, Jaemin Shin, Sung-Ju Lee
*Conference on Neural Information Processing Systems (**NeurIPS**) 2024*

Design Opportunities for Explainable AI Paraphrasing Tools
Yewon Kim, **Thanh-Long V. Le**, Donghwi Kim, Mina Le, Sung-Ju Lee
*Conference on Designing Interactive Systems (**DIS**) 2025*

FlowTool: Controlling Tool Parameter In Image Retouching Via Flow Matching
Thanh-Long V. Le, Steven Walton, Seunghyun Yoon, Branislav Kveton, Trung Bui, Eunho Yang, Viet Lai
Under review at *International Conference on Learning Representations (**ICLR**) 2027*

Beyond Scores with Reward-Model Embeddings for Diverse Flow RL
Yunfei Xie, Yidong Huang, **Thanh-Long V. Le**, Chen Wei, Anup Rao, Viet Lai, Subhojyoti Mukherjee, Branislav Kveton
Under review at *International Conference on Learning Representations (**ICLR**) 2027*

Work Experience

Luxoft – BMW Korea **Mar. 2024 – Dec. 2024**
C++ Software Engineer *Seoul, South Korea*

- Developed and optimized features for map provision application to enhance the accuracy and reliability of ADAS (Advanced Driving Assistance System) functionalities of BMW vehicles.
- Detected, analyzed, and prepared reports and fixes for vehicle anomalies by leveraging software logs and simulations.

Coc Coc Company Ltd. **Sep. 2021 – Oct. 2022**
Software Engineer *Hanoi, Vietnam*

- Maintained the functionality of three key components of Coc Coc Search's backend system – compositor, knowledge graph, and search engine.
- Enhanced the quality of Coc Coc's Answer Box by implementing multilayer rule-based question-answering system and incorporating various alternative logic to address common query patterns, resulting in a 200% increase in impression and a 30% increase in click-through rate.
- Integrated Prometheus into services to enable automatic recording of metrics, eliminating the need for manual retrieval via logs.
- Conducted a full codebase refactoring of compositor system to eliminate potential bugs and technical debt.
- Utilized SQL to collect and analyze usage data of Coc Coc Translate over the course of a year, providing insights to assist managers in decision making.

Honors and Awards

Qualcomm-KAIST Innovation Award (2023): Winner.
KAIST Dean's List (2020): Top 3% students of the department.
KAIST International Undergraduate Scholarship (2019): Full-ride scholar.
Deakin Vice-Chancellor's International Scholarship (2019): Recipient.
Vietnam National Mathematical Olympiad (2018): Silver medal.

Technical Skills

Languages: C++, C, Python, Java, Go, SQL, HTML, CSS, JavaScript.
Technologies/Frameworks: PyTorch, verl, TRL, Unsloth, Jax, Haliax, Levanter, Linux, Git, Vim, Bazel, Bash Scripting.
Certificates: 8.0 in IELTS, 1500 in SAT, 800 in SAT Math Level 2, 800 in SAT Physics.

Academic Services

Conference Reviewer: CHI 2024, ICML 2026, NeurIPS 2026, EMNLP 2026, ICLR 2027.